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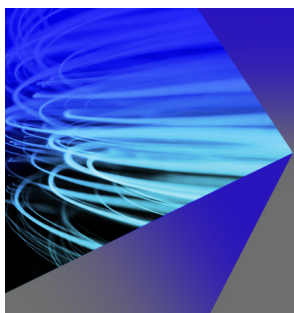
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ABSTRACT

The effect of delta-tabs with its apex leaning downstream on the jet mixing characteristics of a Mach 1.5 elliptic jet of aspect ratio (AR) 3.37 is reported in this paper. The delta-tabs were placed along the minor and major axes of the elliptic nozzle, referred to as tab configuration C1 and C2, respectively. For each tab configuration, four different orientation angles varying from $\theta = 90^\circ$ to 150° were studied, where $\theta = 90^\circ$ is the tab positioned normal to the nozzle exit and $\theta = 150^\circ$ is the tab oriented in the downstream direction at the nozzle exit. The tab effects on jet manipulation were investigated via shadowgraph visualization, and the mean Pitot pressure was measured along the jet centerline and along the orthogonal planes of the jet. The non-dimensionalized supersonic core length L_c^* , average shock-cell spacing $\bar{L}_s^*$, and the axis-switching location x_{as}^* were examined in detail to understand the jet characteristics of the natural and manipulated jets. The mixing capability of the manipulated jets quantified based on the reduction in supersonic core length ΔL_c^* depends strongly on the ideally expanded jet Mach number M_j , tab configuration (C1 and C2), and θ . As such, the ΔL_c^* for the manipulated jet at $\theta = 135^\circ$ is the largest, revealing the optimal tab orientation for jet mixing enhancement, as also observed for the circular jet [Zaman *et al.*, “Control of an axisymmetric jet using vortex generators,” Phys. Fluids **6**(2), 778–793 (1994)]. The relationship between ΔL_c^* and ΔL_s^* is presented to understand the correlation between the jet mixing and noise characteristics of the manipulated jet. Discussion is conducted based on the dependence of ΔL_c^* and ΔL_s^* with θ , which provides useful insight into the jet flow physics. With suitable placement of tabs C1 and C2, the axis switching could be either stopped or augmented. The superior mixing characteristics of an elliptic jet relative to a circular jet, which were found in previous works on the supersonic over-expanded jet [Kumar and Rathakrishnan, “Characteristics of a supersonic elliptic jet,” Aeronaut. J. **120**(1225), 495 (2016)] prevail in the supersonic jet under-expanded jet as well.

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NOMENCLATURE

a	length of semi-major axis
b	length of the semi-minor axis
D	equivalent diameter of the elliptic nozzle
h	height of the delta-tab
L_c^*	non-dimensionalized supersonic core length
$\bar{L}_s^*$	non-dimensionalized average shock-cell length
M_d	nozzle design Mach number
M_j	ideally expanded jet Mach number
NPR	nozzle pressure ratio ($=p_{0s}/p_a$)
p_a	atmospheric pressure
p_e	nozzle exit pressure
p_{0s}	settling chamber pressure
p_{0t}	pressure measured by the Pitot tube
Re_j	Reynolds number based on ρ_j , U_j , D , and μ_j

T_0	temperature in the settling chamber
U_j	jet exit velocity
w	width of the delta-tab
x	coordinate along the jet axis
y	coordinate along the minor axis of the elliptic nozzle
z	coordinate along the major axis of the elliptic nozzle
ρ_j	density of the jet
μ_j	dynamic viscosity of the jet

Superscript

* normalization by D

Subscript

j jet exit conditions

Abbreviation

AR aspect ratio ($=a/b$)

I. INTRODUCTION

Jet control implies the modification of flow and noise characteristics of a jet to meet specific needs. Because of its wide range of application potential, control of free jets aiming at enhancing its mixing with the surround air to which it is discharged has been an active area of research. A concise survey on the various jet control techniques for jet mixing has been presented in open literature.¹ Tabs have been found to be one of the effective means to promote jet mixing.² Bradbury and Khadem³ were the first to investigate the effect of the tab on jet control. They noted that the insertion of small rectangular tabs at the nozzle exit had a profound effect on reducing the potential core length to about two diameters, followed by a rapid decay of the centerline mean velocity. They further speculated that the small-scale vortices shed by the tab was responsible for the mixing promotion. This speculation was later confirmed by the experiments of Zaman *et al.*,⁴ who showed that the tab produced a pair of counter-rotating streamwise vortices. This research^{2–4} has established that the tabs introduce streamwise vortices to transport the low-speed fluid entrained at the jet periphery toward the centerline while forcing out higher speed core fluid. Two sources have been found to be responsible for the generation of the streamwise vorticity by the tab.⁴ The first source S1 is the pressure hill formed at the face of the tab. The second source S2, the pressure gradients formed on the tab surface, leads to vortex shedding from the sides of the tab. The cumulative strength of the vortices formed from the source S1 and S2 depends on the tab orientation. As these vortices are essentially transverse while leaving the tab, they reorient to become streamwise soon after leaving the tab.¹ A detailed investigation by Zaman *et al.*⁴ on the effect of tab shape on a Mach 1.8 jet revealed that the triangular-shaped tab, referred to as a delta-tab, is found to be the most effective in producing the streamwise vortices, with a consequential large influence on the overall jet evolution than its counterpart, the rectangular tab. Recently, Kumar and Rathakrishnan^{5–7} also reported that the triangular-shaped tab is found to be the most effective in jet mixing enhancement than the rectangular tabs. They did argue the effectiveness of the mixing promoting small-scale vortices shed by triangular tab in promoting the mixing of a supersonic jet in the pressure of adverse, zero, and favorable pressure gradient.

An important aspect which is found to be congenial for rapid mixing of the jet with the surrounding fluid is the elliptic shape of the nozzle.^{8,9} The entrainment ratio of an elliptic jet was several times greater than that of a circular jet.^{10,11} Apparently, the mixing promoting vortices shed from an elliptic nozzle will have a continuously varying size over each quadrant. The largest among them will be at the extremities of the minor axis and the smallest at the extremities of the major axis.^{7,12–14} As a result of this, the entrainment occurs around the minor-axis plane as the vortex core moves away from the jet axis and the external fluid is entrained into the jet stream.^{11,15} Thus, to explore differential entrainment along the minor and major axes of the elliptic jet, Kumar and Rathakrishnan¹² investigated the effect of nozzle aspect ratio *AR* on the mixing characteristics of Mach 2 elliptic free jet in the presence of adverse and marginally favorable pressure

gradients at the nozzle exit. They documented that the *AR* 4 jet features better mixing than the *AR* 2 and *AR* 3 jets at all nozzle pressure ratios. Later, Kumar and Rathakrishnan⁷ coupled the effect of the elliptic jet and tab to study the jet mixing characteristics of a Mach 2.0 elliptic jet of *AR* 3 manipulated using rectangular and triangular tabs. This study revealed that tabs were once again proved to be a better passive control than the natural elliptic jet. In the recent study on the effect of shifted tabs in the downstream direction of the nozzle exit of a Mach 2.0 jet, Kumar and Rathakrishnan²⁰ argued that the shifted tabs drastically modify the mixing characteristics of the jet.¹⁹ As such, the thrust loss due to the shifted tabs was also reduced. However, the drawback of the shifted tab is its practical fixation at the nozzle for full-scale applications. Thus, the effect of the shifted tab may be brought in by orienting the tab in the downstream direction, while at the same time the tab can be fixed at the nozzle exit. Perhaps Zaman *et al.*^{4,21} were the first to carry out a detailed study on the tab orientated in the upstream and downstream directions of the nozzle exit. The authors argued that when the apex of the delta-tab was tilted downstream it appeared to work even better than the delta-tab fixed normal to the jet flow and was also preferred from the thrust loss point of view.

The present work is motivated by the fact that the tilted-tab^{4,21} and high *AR* elliptic jets^{10,11,13,16} have proved to be one of the effective passive control techniques in jet mixing enhancement when addressed separately. To the best of the authors' knowledge, there is no study reported in open literature on the effect of the tilted-tab, more specifically the tilted-delta-tabs on elliptic jets in quantifying the jet mixing characteristics. Naturally, one may wonder whether the tilted-delta-tab is a potential passive control technique for increasing the jet mixing of an elliptic jet. There are two potential benefits if the tabs are tilted in the downstream direction. First, the thrust loss associated with the tab can be drastically reduced.^{4,22} Second, the base drag of the tab could also reduce if the tabs are tilted in the downstream direction.²⁰ Now one may also wonder what is the best location for placing the tilted-delta-tab and whether the tabs should be placed along the minor or major axis of the elliptic jet. Indeed previous work reveals that the tab employed at the minor axis has the maximized jet mixing as compared to the tabs placed along the major axis.^{7,14} However, those works were carried out for the tab oriented at $\theta = 90^\circ$ to the jet flow. As such, it would be interesting to see the effect of tab orientation on jet mixing for the tilted-tab configuration.

In order to address the above issue, the present work embarks on a systematic experimental study of an elliptic jet manipulated by a tilted-delta-tab. The delta-tabs were placed along the minor and major axes of the elliptic jet, referred to as tab configuration *C1* and *C2*, respectively. For each tab configuration, four different tab orientations are tested, viz., tabs mounted normal to the flow direction ($\theta = 90^\circ$) to the tabs oriented in the downstream direction $\theta = 120^\circ$, 135° , and 150° (see Fig. 2). Extensive Pitot measurements were performed along the jet centerline, and the jet stream has been visualized using the shadowgraph technique in the orthogonal planes of the manipulated jet. The paper is organized as follows. Experimental setup and measurement details are given in Sec. II, followed by the discussion of mixing characteristics of the natural jet and manipulated jet in Secs. III and IV, respectively. Notable parameters discussed in Secs. III and IV are the non-dimensionalized supersonic core length L_c^* , average shock-cell length $\bar{L}_s^*$ and the axis-switching location x_{as}^* . Finally, the present work is concluded in Sec. V.

II. EXPERIMENTAL DETAILS

A. Jet facility and experimental model

The experiments for the present study were carried out in the open jet facility at High-Speed Aerodynamics Laboratory, Indian Institute of Technology Kanpur. A schematic sketch of the open jet facility is shown in Fig. 1(a). See Refs. 23–25 for details regarding the jet facility. Here we briefly discuss the important features of the jet rig. The compressed air at desired pressure from storage tanks is delivered to the settling chamber through a pressure-regulating valve. The required nozzle pressure ratio (NPR) is maintained by controlling the settling chamber pressure (p_{0s}) since the NPR is the ratio of the settling

chamber pressure (p_{0s}) to the pressure of the ambient (p_a) to which the jet is discharged. Pressure tapings are provided in the settling chamber to monitor the stagnation pressure p_{0s} . The maximum chamber pressure that one may achieve is about 9 bar. The test models are fixed at the end of the settling chamber by a slot holder arrangement [Fig. 1(a)], which is a short pipe like protrusion with an embedded O-ring to prevent leakage.

An elliptic nozzle of Mach 1.5 as shown in Fig. 1(b) was used in this study. The choice of Mach 1.5 jet is explained as follows. First, most of the studies reported in open literature to document the jet mixing and jet acoustics characteristics fall within the range of Mach 1.5 and 1.8 circular jet.^{4,26–28} Thus, the present investigation facilitates

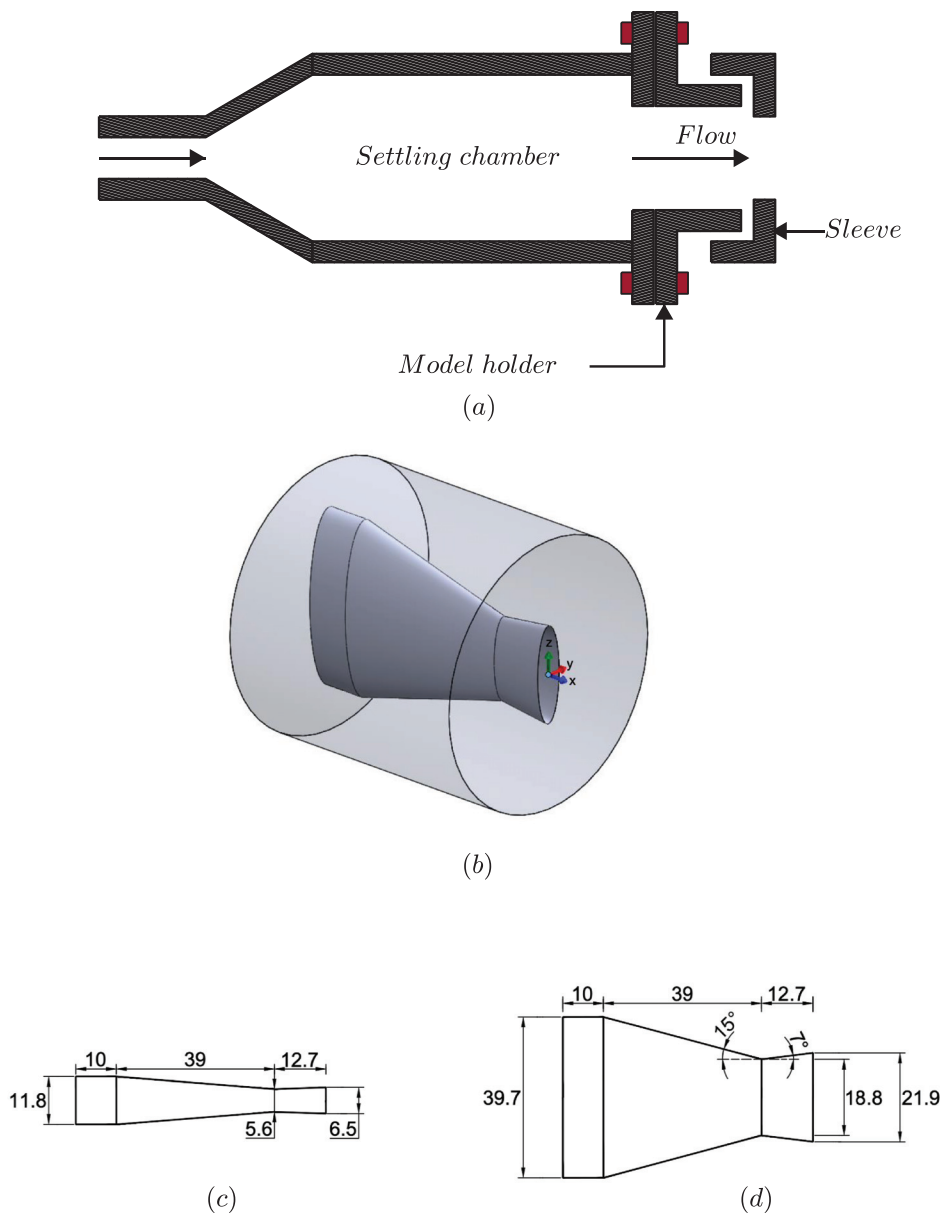


FIG. 1. (a) Schematic drawing of the open jet facility, (b) 3D image of the elliptic nozzle, (c) cut-view of the elliptic nozzle along the minor axis ($x - y$ plane), (d) cut-view of the elliptic nozzle along the major axis ($x - z$ plane). All dimensions are in mm, figure not to scale.

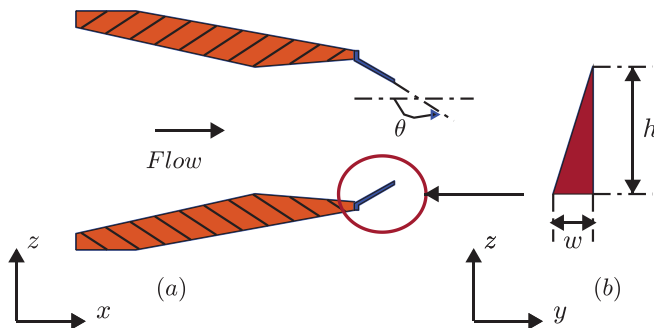


FIG. 2. (a) Tabs positioned at the nozzle exit and (b) tab dimension.

in comparing the behavior of elliptic jet with that of the circular jet reported in open literature. Second, the data available in jet literature for a supersonic elliptic jet are mostly concentrated in the over-expanded and perfectly expanded states of the jet.^{12,13} As such the data on jet studies for under-expanded supersonic elliptic jet are very scarce or not available. However, there have been few studies reported in open literature on under-expanded sonic elliptic slot jet^{29,31} and not on under-expanded supersonic elliptic jet. Hence, a Mach 1.5 elliptic jet was chosen, such that $NPR \geq 3.67$ delivers an under-expanded condition. Kumar and Rathakrishnan¹² performed jet mixing studies on a Mach 2.0 elliptic jet with varying $AR = 2-4$, for $NPR = 4-8$. Their experimental conditions mostly fall in the over-expanded condition of the Mach 2.0 elliptic jet. They argued that the jet mixing for $AR = 4$ was substantially greater than $AR = 2$ and 3 , at all NPR . However, the jet mixing characteristics of a supersonic elliptic that fall between $AR = 2-3$ and $3-4$ is not explored. Hence, the present experimental program consists of a series of AR between 3 and 4 , namely, 3.4 and 3.7 to study the mixing characteristics of a Mach 1.5 elliptic jet. Hence two different nozzle of $AR = 3.4$ and 3.7 are fabricated, and the results of $AR = 3.4$ are reported in the present work. However, after fabrication, the estimated AR was found to be 3.37 . It is worth mentioning that no change in design Mach number of the nozzle was observed after calibration. Thus, in the present investigation, a Mach 1.5 elliptic nozzle of $AR = 3.37$ was chosen to study the jet mixing characteristics.

The nozzle used in most of the previous studies on elliptic jet was either an elliptic orifice or slot^{29,30,32} or a nozzle of circular inlet contouring to an elliptic exit.^{11,16,33} Indeed, these nozzles do not represent the actual characteristics of a jet issuing from an elliptic nozzle. Interestingly, Lee and Baek³² compared the mean streamwise velocity decay of contoured elliptic nozzle¹¹ and orifice type elliptic nozzle.³³ The orifice type or sharp-edged elliptic jets have a larger mean streamwise velocity decay than contoured nozzle jets.^{32,34} Thereby the orifice nozzle has an extremely thin initial boundary layer, which produces strong azimuthal variations in the induced velocity and large entrainment before the vortical structure is diffused by viscosity.¹⁶ Thus, a nozzle with elliptic cross section from inlet to exit is fabricated for the present investigation. In other words, the cross section of convergent-divergent (CD) nozzle is elliptic throughout the nozzle.

Classical research on supersonic jet noise has focused on studying a smoothly varying method of characteristic type of converging-diverging nozzle. Practical nozzles tend to have relatively sharp throats at the intersection of two conical sections. This sharp throat can cause

strong internal shocks that adversely affect nozzle performance and generate shock noise components across the jet operating range. Cuppoletti *et al.*³⁵ investigated the effect of nozzle contour on the thrust and noise characteristics of a Mach 1.61 jet. The Mach 1.61 jet was issued from a converging-diverging nozzle with a sharp and contoured throat, to examine the effect of nozzle contour. This study reveals that at equivalent nozzle pressure ratios, the contoured nozzle provides 10% higher thrust with no increase in mixing noise or shock noise. In fact, the present study is concerned mainly on jet mixing characteristics. Thus, in the present study, we use a conical converging-diverging nozzle with a sharp throat rather than a contoured throat. The equivalent throat (D^*) and exit diameter (D_e) of the nozzle are 11 ± 0.01 and 11.93 ± 0.01 , respectively. This corresponds to the nozzle area ratio (A_e/A^*) of 1.176, such that the isentropic nozzle design Mach number is 1.5. The equivalent diameter $D_e = \sqrt{4ab}$, where a and b are semi-major and minor axis dimensions, respectively. Using the equivalent diameter D_e for normalization means that the comparisons will, in general, be at constant mass flow rate. This facilitates in comparing the jet characteristic of a circular and elliptic jet at identical M_d and NPR or M_j . The length of major and minor axes of the nozzle exit is 21.9 ± 0.01 and 6.5 ± 0.01 mm, respectively, such that the AR is 3.37. Following Kumar and Rathakrishnan,⁵ the semi-divergence angle of the CD nozzle is 7° , and the corresponding expansion section is about 12.7 ± 0.01 mm in length.

Calibration of the nozzle is important since there is a possibility of actual Mach number to differ from the design Mach number. Therefore, the Pitot probe was moved from one end of the nozzle exit to the other end along the major and minor axis for different NPR s. Following Kumar and Rathakrishnan,⁶ the Mach number at the nozzle exit was calculated from the measured Pitot pressure using the normal shock relation. That is, the measured p_{0t} can be used to determine the design Mach number M_d from the known settling chamber pressure p_{0s} using Eq. (1). This shows the nozzle delivers a fairly uniform Mach number at all operating NPR s (figure not shown)

$$\frac{p_{0t}}{p_{0s}} = \left(1 + \frac{2\gamma}{\gamma-1}(M_d^2 - 1)\right)^{\frac{\gamma}{\gamma-1}} \left(\frac{(\gamma+1)M_d^2}{(\gamma-1)M_d^2 + 2}\right)^{\frac{1}{\gamma-1}} \quad (1)$$

Following Zaman *et al.*⁴ and Kumar and Rathakrishnan,⁶ two sets of triangular tabs were designed for the experiments and placed diametrically opposite to each other at the nozzle exit via screws. Indeed, it has been argued by many researchers that the reduction in potential core and growth of characteristics decay for 2 tabs configuration is substantially greater than 4 and 8 tabs.³ Hence, for the present study, two tabs located at diametrically opposite ends of the nozzle have been studied. Zaman¹⁸ documented that the optimum increase in the jet entrainment with minimum thrust loss occurred for a size that corresponded to an area blockage of about 1.5%–2.5% per tab. Thus, we use the tab blockage of 2.5% per tab. A detailed investigation on the shape of tab by Zaman *et al.*⁴ and Kumar and Rathakrishnan⁵⁻⁷ reveals that the delta-tab is found to be the most effective one in increasing the jet mixing than the rectangular tab. Following Kumar *et al.*,^{7,14} the height of the tabs was kept constant at 2 mm. This facilitates in comparing the results of the present investigation with that of Mach 1.5 elliptic jet manipulated by rectangular and triangular tabs reported in literature.^{7,14} Given the tab blockage of 2.5% and height of 2 mm, the width of the delta-tabs was estimated to be 2.8 mm. The

tabs used in the present investigation are made of 0.5 mm thick brass strips. In fact, Zaman *et al.*^{4,21} investigated the effect of tab orientation on the mixing and noise characteristics of a Mach 1.8 circular jet manipulated by delta-tabs. In their study, three different tab orientations were studied, viz., tab tilted in the upstream direction ($\theta = 45^\circ$), tab normal to the jet flow ($\theta = 90^\circ$), and tab tilted in the downstream direction ($\theta = 135^\circ$). They argued that the tab tilted in the downstream direction ($\theta = 135^\circ$) leads to an increase in mixing than the other tab configurations ($\theta = 45^\circ$ and 90°). As a matter of fact, one may wonder what would be the mixing characteristics of the tilted-delta tab for $\theta = 90^\circ$ – 135° and $\theta > 135^\circ$, say $\theta = 120^\circ$ and 150° . Would the optimum tab orientation $\theta = 135^\circ$ remain the same even for an elliptic jet. Thus, to characterize the effect of tab orientation on the mixing characteristics of an elliptic jet, four different tab orientations were tested, viz., tabs mounted normal to the flow direction ($\theta = 90^\circ$) to the tabs oriented in the downstream direction $\theta = 120^\circ$, 135° , and 150° (see Fig. 2). It is important to note that as the tabs are oriented in the downstream direction, the effective blockage of the tabs reduces and, hence, the thrust loss reduces.²⁰ As such, we present in Table I the effective blockage of various tilted-delta-tabs in the present study. However, in the present study, there has been no attempt made to quantify the thrust loss for various tab orientation.

B. Flow measurement and data processing

The pressure in the jet field was measured using a Pitot tube of 0.4 mm inner diameter and 0.6 mm outer diameter, fixed to a 3D traverse with a resolution of 0.1 mm in linear translation. Pressures were measured with a 16 channel Pressure Systems, Inc. 9010 transducer with a range of 0–2.1 MPa. The pressures were recorded at a sampling frequency of 250 Hz, for a time duration of 60 s, with a repeatability within $\pm 2\%$. To visualize the shock-cell structure in a supersonic jet is, the shadowgraph technique was used with a helium spark arc light source in conjunction with a concave mirror of 150 mm diameter. The light source was collimated by the condenser lens and was then brought to the concave mirror. The parallel beam from the mirror was made to pass through the jet flow field and projected on the screen. The surface finish of the mirror is $\lambda/6$ and the focal length is 1.6 m. See Refs. 6 and 24 for further additional details on the instrumentation.

The coordinate system (x, y, z) is defined in Fig. 2. Measurements were conducted along the major axis ($x - z$) plane and along the minor axis ($x - y$) plane of the jet. Note that x^* , y^* , and z^* are the non-dimensionalized coordinate of x , y , and z , respectively by D_e . Experiments were conducted for the jet Reynolds number $Re_j = 0.42 \times 10^5$ and 1.12×10^5 , respectively with an uncertainty less than $\pm 1\%$ for the minimum and maximum M_j of the present investigation (Table II). The nozzle pressure ratios used, the corresponding ideally expanded jet Mach number M_j , expansion ratio p_e/p_a , and

TABLE I. TABLE effective blockage.

θ	Effective blockage (%)
90°	5.0
120°	4.4
135°	3.4
150°	2.5

TABLE II. Test conditions.

NPR	M_j	$Re_j (*10^5)$	p_e/p_a
3	1.36	0.42	0.82
4	1.56	0.64	1.09
5	1.71	0.71	1.36
6	1.83	0.92	1.63
7	1.93	1.01	1.91
8	2.01	1.12	2.18

other parameters of interest are shown in Table II. The settling chamber pressure (p_{0s}) was maintained constant during the experiments with an uncertainty less than $\pm 1\%$. The experimental uncertainty of L_c^* , $\bar{L}_s^*$, and x_{as}^* was determined from repeated measurements and calibration of Pitot tube. For the indirectly measured quantities such as ΔL_c^* , the method of propagation of uncertainties is used to determine their error bars, and the maximum uncertainties of ΔL_c^* was found to be less than $\pm 8\%$.

III. FLOW STRUCTURE OF THE NATURAL JET

The purpose of this study was to understand the mechanism behind the jet mixing characteristics of natural and manipulated jets. The prime features of the jet, such as the non-dimensionalized supersonic core length L_c^* , average shock-cell length $\bar{L}_s^*$, and decay in the characteristic region are the parameters chosen to characterize jet mixing.^{3,25} As such, the strong noise sources for a jet were distributed within the first ten diameters of the jet flow.²⁶ Indeed, the third and fourth shock-cells are the source for broadband shock noise.²⁰ Interestingly, the L_c^* , $\bar{L}_s^*$, and characteristic decay can be discerned from the measured Pitot pressure along the jet centerline. Shorter L_c^* and $\bar{L}_s^*$ and faster decay in the characteristic decay region are an indication of rapid mixing of the jet. However, in this paper, the jet mixing quantification is quantified in terms of L_c^* and $\bar{L}_s^*$.

It is important to note that, in a supersonic jet core, the Pitot probe measures the total pressure behind the bow-shock that stands ahead of the probe nose and not the actual total pressure. If the actual total pressure is required, then one has to correct the measured pressure for the pressure loss across the shock, which is practically impossible because of the complex shock-cell structure in the jet field. Even a conical supersonic probe will measure only the Mach number downstream of a shock that stands ahead of the probe nose and not the actual Mach number.²⁵ Nevertheless, the difference between the measured Pitot pressure and the actual ones is about less than 5%.³⁶ In fact, the qualitative behavior of the jet stream measured in-terms of either Mach number or Pitot pressure would remain the same.^{37,38} In other words, the jet parameters, namely, the core length, average shock-cell length, characteristic decay, and the full-developed regions of the jet would be the same, even if the jet parameters were quantified in-terms of either Mach number or Pitot pressure.¹ As a matter of fact, the present study is largely focused on characterizing the core length and average shock-cell length of the jet, which can be easily quantified using the Pitot pressure measurements. Hence, the Pitot pressure measurements are accurate enough to capture the overall features of the jet.^{5,6} Thus, the Pitot pressure was measured along the jet centerline at intervals of 1 mm up to $20D_e$ to capture L_c^* and $\bar{L}_s^*$ and the

characteristic decay. Figure 3 represents the variation of p_{0t}/p_{0s} with x^* for the Mach 1.5 elliptic jet at various M_j . For reference, the p_{0t}/p_{0s} variation of a Mach 1.5 circular jet^{26,36} at relatively identical M_j is also presented in Fig. 3. As such, the shadowgraph images captured along the major ($x-z$ plane) and minor ($x-z$ plane) axes of the natural elliptic jet at various M_j are shown in Fig. 4 for comparison. In fact, it is worth noting that the Pitot pressure oscillations in Fig. 3 represent the presence of a shock-cell structure in the jet field as clearly seen from the shadowgraph visualization (Fig. 4). Following Kumar and Rathakrishnan,⁶ the L_c^* was estimated from the centerline Pitot pressure measurements and is used to quantify the level of mixing of the jet with the ambient fluid. The supersonic core length L_c^* is defined as the distance from the nozzle exit to the axial location along the jet centerline, where the local flow Mach number is equal to 1.0. However, the L_c^* estimated from centerline Pitot pressure is briefly presented here. It is well known that for air $\gamma = 1.4$, so the isentropic pressure

ratio p_{0t}/p_a at which the sonic flow ($M_j = 1$) occurs as estimated from Eq. (2) is 1.89^{1,6}

$$p_{0t}/p_a = \left(1 + \frac{\gamma-1}{2} M_j^2\right)^{\frac{\gamma}{\gamma-1}}. \quad (2)$$

Thus, the cutoff Pitot pressure for sonic Mach number $(p_{0t}/p_{0s})_{M=1}$ is given by as follows:

$$(p_{0t}/p_{0s})_{M=1} = \frac{1.89}{p_{0s}/p_a}. \quad (3)$$

Thus, for any given p_{0s}/p_a or NPR, the $(p_{0t}/p_{0s})_{M=1}$ can be estimated from Eq. (3). For example, at $p_{0s}/p_a = 3$, the $(p_{0t}/p_{0s})_{M=1}$ estimated from Eq. (3) is 0.63. Hence, the cutoff Pitot pressure $(p_{0t}/p_{0s})_{M=1}$ of 0.63 is marked in Fig. 3(a1). See Table II for the corresponding p_e/p_a for each p_{0s}/p_a or NPR. As such, similar $(p_{0t}/p_{0s})_{M=1}$ for other p_{0s}/p_a is

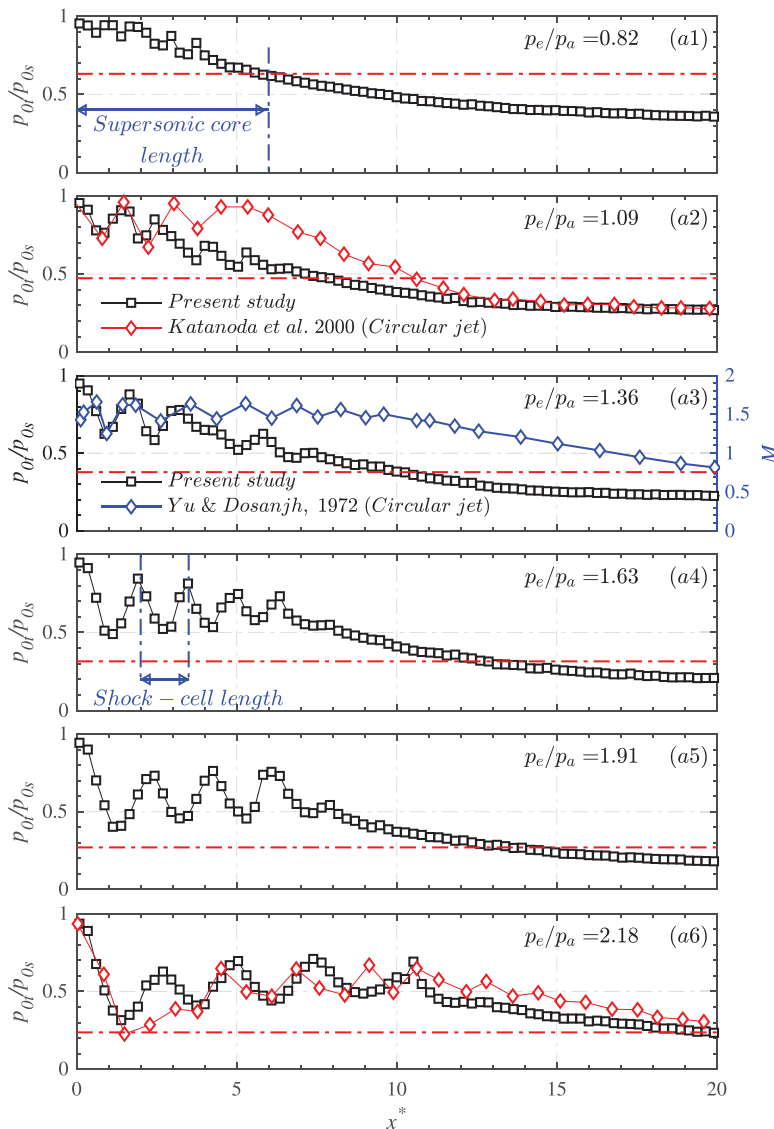


FIG. 3. Centerline Pitot pressure decay for the natural elliptic jet for various p_e/p_a : (a1) $p_e/p_a = 0.82$, (a2) $p_e/p_a = 1.09$, (a3) $p_e/p_a = 1.36$, (a4) $p_e/p_a = 1.63$, (a5) $p_e/p_a = 1.91$, and (a6) $p_e/p_a = 2.18$. The dotted line here represent the $(p_{0t}/p_{0s})_{M=1}$ as given by Eqs. (3) and (4).

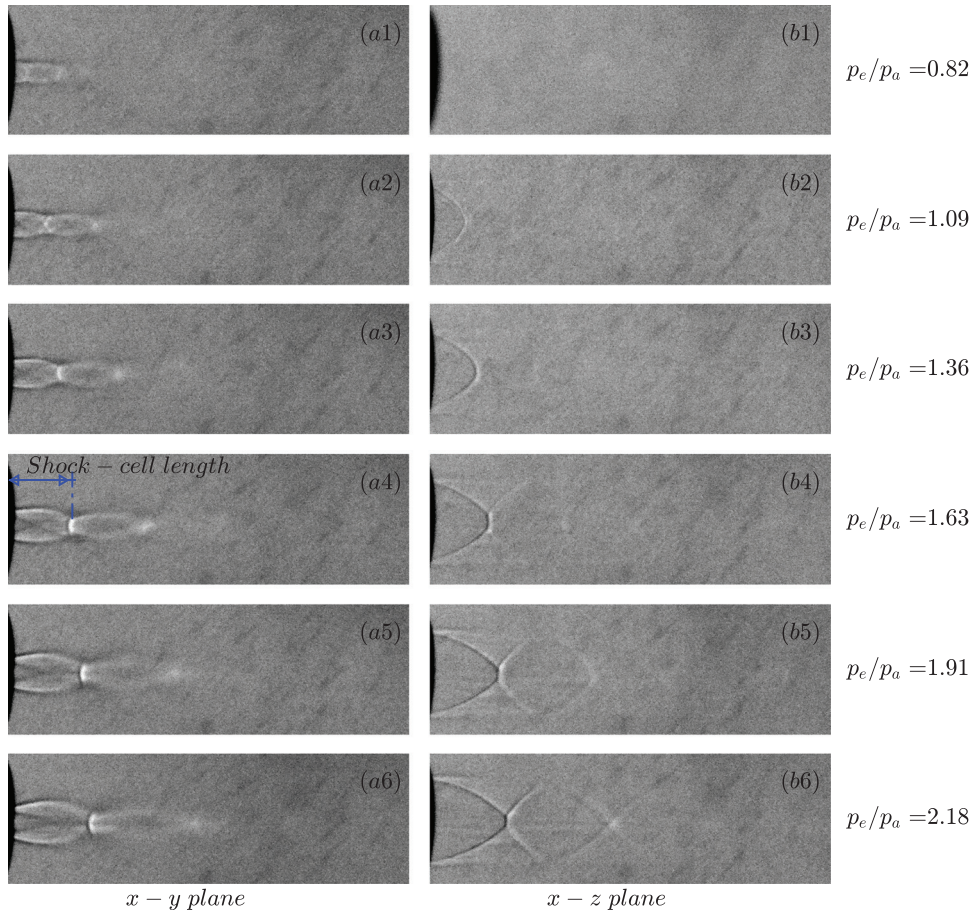


FIG. 4. Shadowgraph images captured along the $x - y$ and $x - z$ planes of the natural elliptic jet for various p_e/p_a : [(a₁)–(b₁)] $p_e/p_a = 0.82$, [(a₂)–(b₂)] $p_e/p_a = 1.09$, [(a₃)–(b₃)] $p_e/p_a = 1.36$, [(a₄)–(b₄)] $p_e/p_a = 1.63$, [(a₅)–(b₅)] $p_e/p_a = 1.91$, [(a₆)–(b₆)] $p_e/p_a = 2.18$.

estimated and is marked in Figs. 3(a2)–3(a6). The x^* corresponding to $(p_{0t}/p_{0s})_{M=1}$ is identified as the length of supersonic jet core or L_c^* . This highlights that the flow below this cutoff line is subsonic. Further, the Pitot pressure oscillations are a direct measure of strength of the shock-cell structure in the jet field.²⁰ Note that the distance between two successive peaks of Pitot pressure signifies the shock-cell length [Fig. 3(a4)] as also clear seen from the shadowgraph images [Fig. 4(a4)]. The shock cell spacing is nearly uniform over the first four cells. Beyond this, the shock-cells become shorter due to viscous and mixing effects, and hence the thickening of shear layer. Thus, its accurate measurement at the downstream location is sometimes very difficult.^{29,39,40} Indeed, the thickening of shear layer reduces the $\bar{L}_s^*$. In-fact the effective source for noise is located at or downstream of the end of third shock cell.⁴¹ Thus, following Baier *et al.*⁴² and Heeb *et al.*,⁴³ we determine the average shock-cell length $\bar{L}_s^*$ of the first four shock-cells from the measured p_{0t}/p_{0s} . The dependence of L_c^* and $\bar{L}_s^*$ with M_j is presented later. It is very common in jet literature to represent the jet parameters as a function of either M_j or NPR or p_e/p_a . Thus, in the present investigation, we chose M_j as the dependent parameter to understand the jet mixing parameters L_c^* and $\bar{L}_s^*$.

It is worth mentioning that p_e/p_a (marked in Fig. 3) is evaluated based on the following equation as given in Refs. 1 and 6:

$$p_e/p_a = \frac{p_e}{p_{0s}} * \frac{p_{0s}}{p_a}, \quad (4)$$

where p_{0s}/p_e and p_{0s}/p_a are the design and operating nozzle pressure ratios. The p_{0s}/p_e for a Mach 1.5 jet estimated from the isentropic equation [Eq. (5)] is 3.67

$$p_{0s}/p_e = \left(1 + \frac{\gamma - 1}{2} M_d^2\right)^{\frac{\gamma}{\gamma - 1}}. \quad (5)$$

Thus, for a given operating nozzle pressure ratio, the p_e/p_a can be estimated from Eq. (4). As discussed earlier, the present work embarks to study the jet mixing characteristics of an under-expanded supersonic elliptic jet, and hence, a great attention is given to $p_e/p_a > 1$.

Few important observations can be made with respect to Figs. 3 and 4. First, for $p_e/p_a = 0.82$ –2.18, the centerline p_{0t}/p_{0s} distribution shows periodic oscillation due to the presence of periodic shock-cells in the jet core [Figs. 3(a1)–3(a6)]. The strength of each shock-cells reduces as the jet propagates in the downstream direction, as revealed by the reduced p_{0t}/p_{0s} of each shock-cell [Figs. 3(a1)–3(a6)]. This observation is further complemented from the shadowgraph images as seen by the reduced intensity in the streamwise direction [Figs. 4(a1)–4(b6)]. Second, for $p_e/p_a = 1.09$, 1.36, and 2.18, the waves in the elliptic jet stream are considerably weaker than the circular jet, as

suggested by the reduced amplitude of Pitot pressure [Figs. 3(a2), 3(a3), and 3(a6)]. For comparison, the centerline p_{0t}/p_{0s} or Mach number distribution for a circular jet at $M_d = 1.5$ and varying M_j is also presented in Fig. 3. It is worth noting that the centerline Mach number data for circular jet of Yu and Dosanjh²⁶ are presented in Fig. 3(a3), rather than p_{0t}/p_{0s} . According to Norum,⁴⁴ the shape of the nozzle exit geometry greatly influences the nature of downstream development of shocks. Similar observation was also reported in open literature for a Mach 2.0 jet issuing from an elliptic and circular nozzle.¹³ A free circular jet being exhausted from a nozzle will entrain air equally on both sides of its shear layer.⁴² However, for an elliptic jet due to the nature of the nozzle exit geometry, the entrainment would be different all along the sides of the jet.¹⁰ Evidently, the different flow topology is observed along the minor [Figs. 4(a1)–4(a6)] and major axis [Figs. 4(b1)–(b6)] of the natural elliptic jet. Thus, an elliptic jet exhibits a varying growth in shear layer along the sides of the jet. Hence, for a given M_d and M_j , an elliptic jet spreads faster than the circular jet. As a result of this, the strength of shock-cell decreases for the elliptic jet. Indeed, the core length of the elliptic jet is substantially smaller than the circular jet. For example, the L_c^* of elliptic and circular jets at $p_e/p_a = 1.36$ is about 10 and 16, respectively. This highlights the efficient jet mixing of the elliptic jet over the circular jet at the under-expanded condition. As one may would expect, the superior mixing characteristics of an elliptic jet relative to a circular jet in the over-expanded regime is reported in open literature.¹³ Thus, it may be stated that the elliptic jet dominates the circular jet, both in the over-expanded and under-expanded state of a supersonic jet. Finally, for all p_e/p_a , the p_{0t}/p_{0s} gradually decreases once the oscillations are eliminated or suppressed. Now one may wonder why the p_{0t}/p_{0s} gradually decreases even in the supersonic regime ($p_{0t}/p_{0s} > (p_{0t}/p_{0s})_{M=1}$). Note that the region above the dotted line represents the supersonic flow and below represent the subsonic flow. Similar observations were also reported in open literature for a supersonic jet.^{45,46} Apparently, the p_{0t}/p_{0s} decreases gradually in the supersonic regime due to the weak shock waves (see Fig. 9 in Chin *et al.*⁴⁵) Phalnikar *et al.*⁴⁶ experimentally studied the flow behavior of a supersonic jet using Pitot probe measurement and Schlieren visualization, and argued that the Pitot pressure decreases gradually in the presence of weak shock waves (see Fig. 7 in Phalnikar *et al.*⁴⁶)

It is of interest to see the variations in L_c^* with p_e/p_a . Thus we present in Fig. 5 the dependence of L_c^* with p_e/p_a for the elliptic jet. Note that, for a given M_d and NPR, the p_e/p_a and M_j are correlated with each other via the isentropic equation¹ [Eqs. (4) and (5)]. Thus, Fig. 5 also presents M_j in the abscissa in addition to p_e/p_a . For comparison, Fig. 5 also presents the dependence of L_c^* with p_e/p_a for the circular jet reported in open literature.^{26,36} It is worth mentioning that the L_c^* presented here was obtained from Fig. 3. Shorter L_c^* is an indication of rapid mixing of the jet fluid with the ambient fluid.^{5,25} As expected the L_c^* of elliptic jet is smaller than the circular jet due to the rapid jet mixing of elliptic jets. Notably, the L_c^* increases as p_e/p_a increases. The increase in L_c^* with p_e/p_a is primarily due to the increased strength of the shock-cell structure with p_e/p_a .

Another important parameter that is commonly used in jet studies is the non-dimensionalized average shock-cell spacing $\bar{L}_s^*$. Thus, we present in Fig. 6 the dependence of $\bar{L}_s^*$ with M_j for the elliptic jet. For comparison the dependence of $\bar{L}_s^*$ with M_j of $AR = 3.37$ elliptic jet determined from Tam's theory⁴⁷ is also presented in Fig. 6. Tam⁴⁷

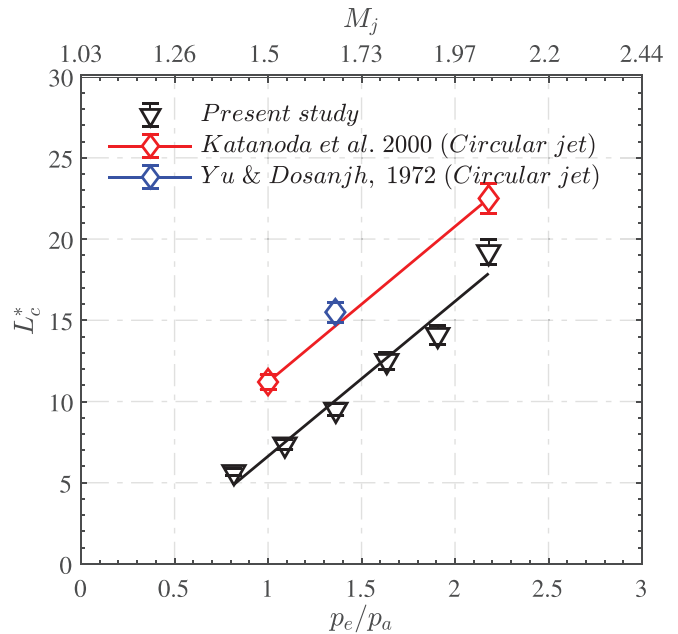


FIG. 5. Dependence of L_c^* with p_e/p_a and M_j for the natural jet at $M_d = 1.5$; elliptic jet—present study, circular jet—Katanoda *et al.*,³⁶ and Yu and Dosanjh.²⁶

argued that a supersonic jet issuing from convergent-divergent nozzles operating at an off design condition experiences the shock-cell structure which may be assumed to be a linear perturbation on the fully expanded condition of the jet. Thus, as given by Tam,⁴⁷ Eq. (6) can be used to determine the $\bar{L}_s$ for a given M_j and known a and b of an elliptic jet, where a and b are the semi-major and semi-minor axis, respectively,

$$L_s = 4b \frac{\sqrt{(M_j^2 - 1)}}{[1 + 1/\pi(b/a)]}. \quad (6)$$

Equation (6) on reducing to non-dimensional average shock-cell length $\bar{L}_s^*$ yields

$$\bar{L}_s^* = 4b^* \frac{\sqrt{(M_j^2 - 1)}}{[1 + 1/\pi(1/AR)]}. \quad (7)$$

For comparison, the dependence of $\bar{L}_s^*$ with M_j for a circular jet predicted from Prandtl-Pack relation⁴⁸ [Eq. (8)] is also presented in Fig. 6

$$\bar{L}_s^* = \pi \frac{\sqrt{(M_j^2 - 1)}}{2.405}. \quad (8)$$

Similar to L_c^* , the $\bar{L}_s^*$ of both elliptic and circular jet increases as M_j increases. As one may would expect, the $\bar{L}_s^*$ of elliptic jet is smaller than the circular jet due to the rapid jet mixing of elliptic jets. Interestingly, a fair amount of agreement between experiment and Tam's theory can be seen for $M_j \leq 1.8$ and significant deviation occurs for $M_j > 1.8$. Indeed, Munday *et al.*⁴⁰ compared the variation of $\bar{L}_s^*$ with M_j of a circular jet predicted from Eq. (8) and experiments for $M_j = 1.0$ –2.0. Apparently, they also observed a reasonably fair collapse between the predictions and measured values of $\bar{L}_s^*$ for lower values

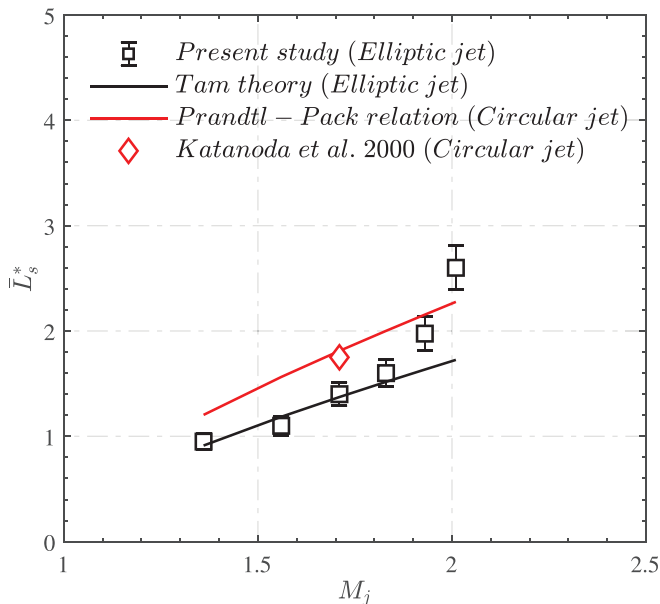


FIG. 6. Dependence of $\bar{L}_s^*$ with M_j for the natural jet.

of M_j , which becomes worse as M_j increases. As such, most of the works reported to validate the theories^{47,48} [Eqs. (6) and (8)] or experimental data²⁹ were limited to $M_j < 1.8$. Hence, the reason why the predicted $\bar{L}_s^*$ deviate significantly from that of experiments for $M_j \geq 1.8$ was not explored in detail. The reason for this may be explained as follows. Tam *et al.*⁴⁹ argued that the linear shock cell model is valid for $M_j < 1.4$ for circular choked jets. As a result of assuming a linear perturbations on the fully expanded condition, the Mach disk occurs immediately downstream of the nozzle exit for $M_j > 1.4$. That is for $M_j > 1.4$ a Mach disk occurs from a fully expanded jet.²³ On the other hand the Mach disk does not occur for a Mach 1.5 jet issuing from a convergent-divergent duct given the $M_j = 1.4$ –2.0 [Figs. 4(a4)–4(b6)]. This results in significant deviation between the experimental work and the predictions $\bar{L}_s^*$. As a matter of fact, the M_j at which the Mach disk occurs for an elliptic jet is greater than that of circular jet. Thus, the predicted $\bar{L}_s^*$ agrees close to that of experiments even for $M_j = 1.4$ –1.8, in contrast to circular jet ($M_j < 1.4$). In other words, the predicted $\bar{L}_s^*$ falls close to the experimental values of a circular jet for $M_j < 1.4$ (Tam *et al.*⁴⁹) and elliptic jet for $M_j < 1.8$. Beyond the limiting value of M_j , the predicted $\bar{L}_s^*$ deviates from the actual $\bar{L}_s^*$ as a result of Mach disk formation in the predicted theories.

It is well known that free-jet undergoes switching of axis due to the differential spread along the major and minor axis, referred to *axis-switching* phenomena. In the last few decades, axis-switching has been observed in various non-circular jets and the mechanism for the same is documented in detail by Gutmark and Grinstein.¹⁰ As a matter of fact, faster axis switching is an indication of increased jet spread and faster near-field mixing of jet with the ambient fluid. To get an insight into the axis switching phenomenon, detailed Pitot pressure measurements were performed along the major and minor axis of the jet at various stream-wise location. The Pitot pressures were measured over the entire major and minor axis without assuming any degree of jet

symmetry. The typical representation of normalized Pitot pressure p_{0t}/p_{0s} measured at various x^* along the y^* and z^* plane of the jet for $M_j = 1.36, 1.83$, and 2.01 is shown in Fig. 7. As one may would expect, at $x^* = 2.0$ and 4.0 for all M_j [Figs. 7(a1)–7(a2), 7(b1)–7(b2), and 7(c1)–7(b2)], the jet spreads faster along the major axis (z^*) than along the minor axis (y^*), owing to the azimuthal asymmetry of the elliptic nozzle exit. It is worth making a note that the occurrence of sharp peaks along the z^* axis [Figs. 7(a1)–7(a2), 7(b1)–7(b2), and 7(c1)–7(b2)] is a result of shock-cell structure in the supersonic jet.^{20,25} In general, as the jet develops in the downstream direction, it spreads through the mixing and entrainment with the surroundings. When the flow in the z^* axis approaches the boundary, the entrainment in the z^* is weakened while the mixing along the y^* axis becomes competitive to the z^* due to the differential shear layer thickness along the z^* and y^* axes. Thus a situation is reached where the spread along both the z^* and y^* are similar, forming a circular cross section. As the jet further develops, the flow along y^* axis exceeds that along the z^* axis [Fig. 7(a4)], which is how the axis-switching occurs. In other words, the initial minor and major axis of the nozzle undergoes a change where the axis is now interchanged owing to the differential spread in y^* and z^* directions. The axial location where the jet spread of both the minor and major axis is similar forming a circular cross section is termed as location of axis-switching x_{as}^* . As such the x_{as}^* for $M_j = 1.36$ is about 7.0 [Fig. 7(a3)]. In other words, the circular cross section of the jet occurs at $x^* = 7.0$ for $M_j = 1.36$ [Fig. 7(a3)]. Similar observation can also be seen for $M_j = 1.83$ and 2.01, where the x_{as}^* is about 8.5 (figure not shown) and 10 [Fig. 7(c3)], respectively.

The variation of x_{as}^* with M_j for the Mach 1.5 elliptic jet at AR 3.37 is shown in Fig. 8. For comparison, the variation of x_{as}^* with M_j for the Mach 2.0 elliptic jet at AR 4 documented by Kumar and Rathakrishnan¹² is also shown in Fig. 8. It is worth noting that x_{as}^* in Fig. 8 is obtained from Fig. 7, where the jet spread of both the minor and major axis is identical. Evidently, the x_{as}^* increases with increase in M_j . Many other researchers have studied and tracked the point of axis switching.^{7,10} It is found that the results obtained in this study are in good agreement with previous work.¹²

In summary, the L_c^* , $\bar{L}_s^*$, and x_{as}^* increases as M_j increases, implying the strength of the shock-cell structures increases as M_j increases. Further, for all M_j , the magnitude of L_c^* and $\bar{L}_s^*$ of the elliptic jet is lowest as compared to the circular jet. This highlights the better jet mixing capability of the elliptic jet than the circular jet.

IV. FLOW STRUCTURES OF THE MANIPULATED JET

In this section, we document the jet mixing capability of an elliptic jet manipulated under tab configuration C1 and C2. As such, two sub-sections are dedicated to understand the mixing capability of tab configurations C1 and C2 on the elliptic jet. Furthermore, for each tab configuration, four different orientation angles varying from $\theta = 90^\circ$ to 150° are also presented.

A. Elliptic jet manipulation-C1

Figure 9 presents the variation of p_{0t}/p_{0s} with x^* of the manipulated jet C1 for $\theta = 90^\circ$ to 150° at $M_j = 1.83$. For reference, the p_{0t}/p_{0s} variation of the natural jet is also shown in Fig. 9. Few important observations can be made. First, for all tab angles $\theta = 90^\circ$ – 150° , the tabs alter and weaken the shock/expansion structure drastically, as clearly seen from the reduced amplitude of Pitot pressure oscillations.

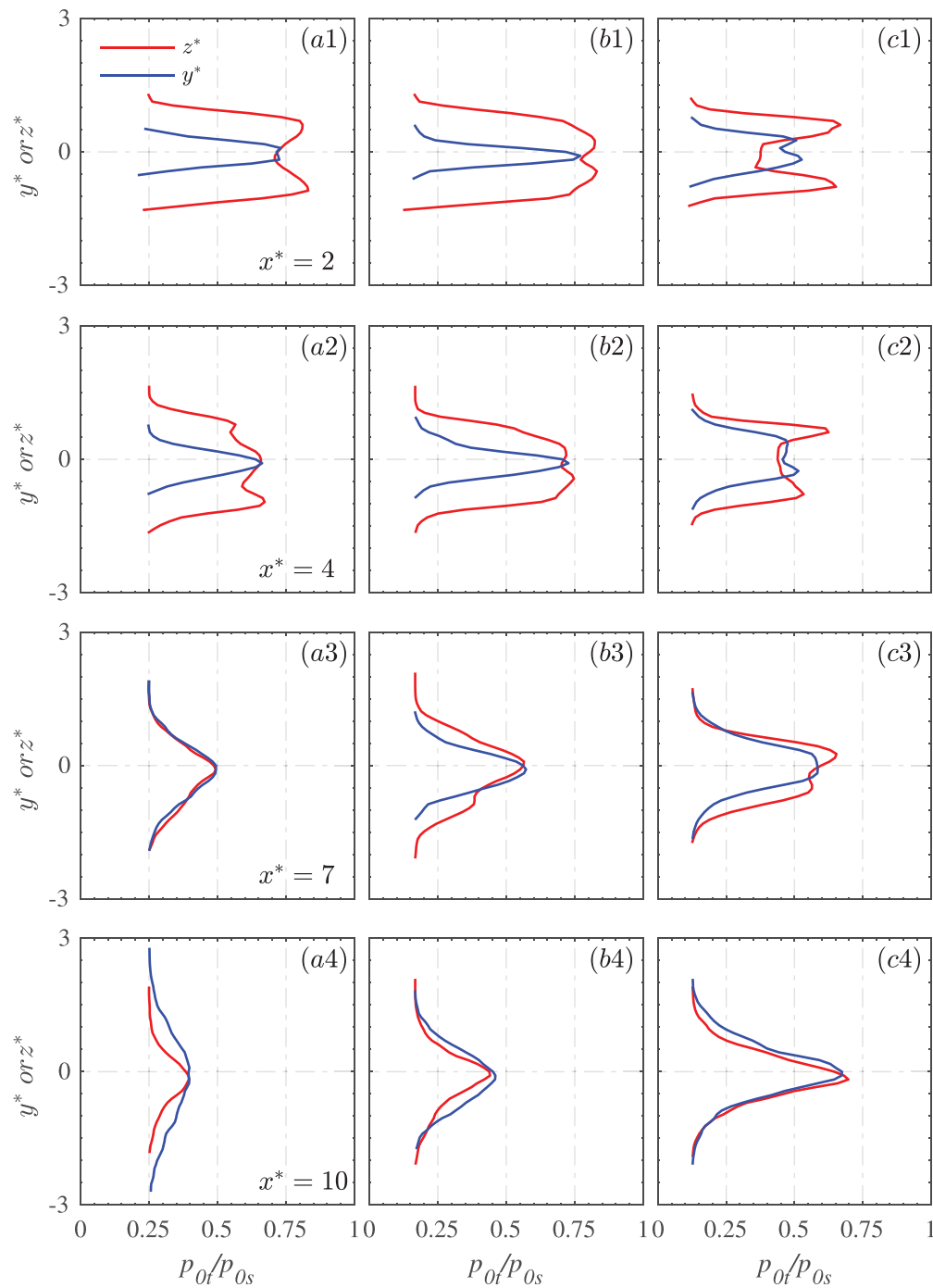


FIG. 7. Pressure profiles at various axial locations of the natural elliptic jet for various M_j ; (a1)–(a4) $M_j = 1.36$, (b1)–(b4) $M_j = 1.83$, (c1)–(c4) $M_j = 2.01$.

The jet centerline Pitot pressure for $\theta = 90^\circ$ – 150° is found to decay much faster than the natural jet and thus jet Mach number, which is generally an indication of increased jet spread. These effects are most pronounced for $\theta \geq 135^\circ$ and the least for $\theta \leq 120^\circ$. Second, for θ

$\geq 135^\circ$, the p_{0t}/p_{0s} below the cutoff pressure $(p_{0t}/p_{0s})_{M=1}$ drops to a minimum and then rises gradually until a maximum (but still less than $(p_{0t}/p_{0s})_{M=1}$) and again drops but gradually to a flat plateau in the far-field region. However, for $\theta = 120^\circ$, the p_{0t}/p_{0s} below the cutoff

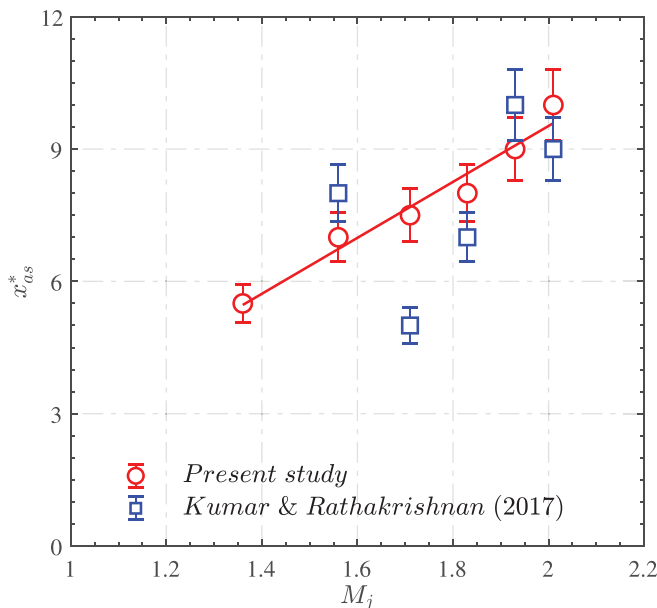


FIG. 8. Dependence of x_{as}^* with M_j for the natural elliptic jet for various M_d .

pressure $(p_{0t}/p_{0s})_{M=1}$ drops and almost remains a flat plateau throughout, but with a slight bump between $x^* = 5$ –10. Many other researchers have investigated the tab effect on jet mixing for the tab placed in the minor axis of an elliptic jet.^{7,13} However, such a variation in p_{0t}/p_{0s} along the jet centerline was not observed in their study. Interestingly, Lovaraju and Rathakrishnan,⁵⁰ studied the jet mixing characteristics of a subsonic jet manipulated by cross-wire (tab that runs along the diameter at the exit) and observed similar p_{0t}/p_{0s} variation along the jet centerline (see Fig. 3 in Ref. 50). As such the iso-Mach contours measured at various streamwise locations reveal that the manipulated jet has undergone jet bifurcation. Thus, to investigate this, we present in Fig. 10 the shadowgraph images of the manipulated jet C1 for $\theta = 90^\circ$ at $M_j = 1.83$ captured along the $x - y$ and $x - z$ planes. As explicitly seen in Fig. 10(a2), the tab completely bifurcates the jet into two streams. This bifurcation causes the jet to behave as if the jet is issuing from a twin jet. That is the behavior of p_{0t}/p_{0s} along the jet centerline for $\theta = 120^\circ$ [Fig. 9(a3)] is similar to that observed for a twin jet along the symmetry line.^{51,52} Thus, the bifurcated jet resembles the characteristics of a twin jet such as merging, converging, and combined regions. Third, it is clear that the p_{0t}/p_{0s} measured along the centerline of the bifurcated jet is in the wake of tab [dotted line marked in Fig. 10(a2)], hence the reduced magnitude of p_{0t}/p_{0s} . Indeed, Zaman *et al.*⁴ also claimed that the bifurcated jet would over-emphasized by the centerline data. Hence, the measured p_{0t}/p_{0s} along the jet centerline of the bifurcated jet may not represent the actual Pitot pressure of a manipulated jet. Finally, one remark is due. Kumar and Rathakrishnan^{7,13} also observed the bifurcation of an elliptic jet when the tabs were placed in the minor axis of the jet (see Fig. 17 in Ref. 13). However, the jet bifurcation observed by Kumar and Rathakrishnan^{7,13} was not a complete bifurcation as observed in the present study. This was clearly evident from the measured Pitot pressure along the jet centerline (see Figs. 4–8 in Ref. 13). A close look into

the nozzle geometry of the present investigation unveils that the length of minor axis ($2b$) is about 6.5 mm and the height h of each tab was about 2 mm; thus, the ratio of tab height to length of semi-minor axis (h/b) is about 0.62. Hence, the tabs employed in the present investigation was almost cutting down the minor axis of the nozzle as similar to a cross-wire or limiting tab. On the other hand, in the work of Kumar and Rathakrishnan,^{7,13} the length of the minor axis was about 9.2 mm and the height of each tab was about 2 mm; thus, h/b was about 0.43. Evidently, the tabs employed in Kumar and Rathakrishnan^{7,13} were not cutting down the entire minor axis of the nozzle. Thus, a partial bifurcation of the elliptic jet was observed in the work of Kumar and Rathakrishnan^{7,13} and a complete bifurcation in the case of present study, respectively. The above observed phenomena are primarily due to the h/b ratio.

In summary, from view of practical significance, it would be not advisable to manipulate a jet so as to behave completely as a twin jet. Thus, employing the tab in the minor axis of the elliptic jet would be governed based on h/b ratio. Presumably for $h/b < 0.5$, the tab may not induce a complete bifurcation of the manipulated jet.

B. Elliptic jet manipulation-C2

Figure 11 presents the variation of p_{0t}/p_{0s} with x^* of the manipulated jet C2 for $\theta = 90^\circ$ – 150° at $M_j = 1.83$. For reference, the p_{0t}/p_{0s} variation of the natural jet is also shown in Fig. 11. Few important observations can be made. First, as observed for manipulated jet C1 (Fig. 9), the manipulated jet C2 at all the tab angles $\theta = 90^\circ$ – 150° weakened the shock/expansion structure. The L_c^* for the natural jet at $M_j = 1.83$ is about 12.5 [Fig. 11(a1)]. Once the tabs are introduced at the nozzle exit, the L_c^* of the manipulated jet reduces considerably. That is the L_c^* for $\theta = 90^\circ$, 120° , 135° , and 150° is about 10.0, 10.5, 7.0, and 7.5, respectively [Figs. 11(a2)–11(a5)]. This highlights the effective manipulation of the elliptic jet under the placement of tabs. In fact, the manipulated jet at $\theta = 135^\circ$ exhibits the shorter L_c^* , implying the maximized jet mixing of all tab angles. Zaman *et al.*⁴ investigated the effect of tab angle on a circular jet Mach number of 1.8. They studied three different tab angles $\theta = 45^\circ$, 90° , and 135° . The $\theta = 45^\circ$ tab was inclined upstream to the nozzle exit and has no significant effect on jet mixing as compared to the other tab orientations. Further, their study reveal that the jet decay rate and the jet mixing were greater for the tab with orientation angle $\theta = 135^\circ$ as compared to $\theta = 45^\circ$ and 90° . Thus, it is found that the results obtained in this study are in good agreement with previous work.⁴ Presumably, the optimum inclination angle for better jet mixing via tab is 135° , irrespective of the nature of the nozzle geometry. The physical reason for the observed phenomena is discussed later. Second, Behrouzi and McGuirk⁵³ studied the effect of varying orientation angle of a jet at $\theta = 30^\circ$, 90° , and 150° . Their study reveals that the potential core length was reduced to around $1D$, $2D$, and $4D$ for tab orientation $\theta = 90^\circ$, 150° , and 30° , respectively. That is, the tab with $\theta = 90^\circ$ orientation angle has the least potential core length. Their study also reveals that orienting the tab in the upstream direction ($\theta = 30^\circ$) has no significant effect on jet mixing as compared to the other tab orientation. Further, they also argued that the tab blockage for $\theta = 90^\circ$ tab was 50% more than for the other two tabs ($\theta = 30^\circ$ and 150°). Thus, the tab with $\theta = 90^\circ$ orientation angle has the least potential core length. Evidently, the increased jet mixing or the least potential core length for the $\theta = 90^\circ$ tab is primarily due to the higher tab blockage, rather than

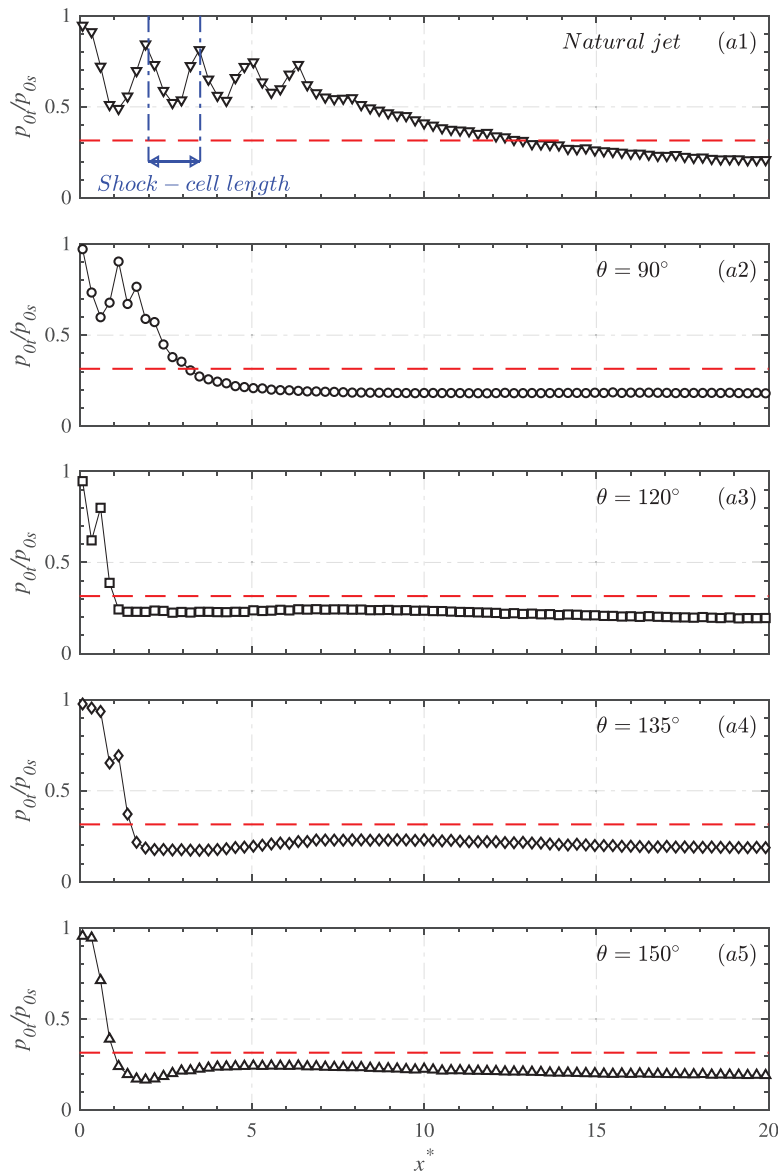


FIG. 9. Centerline Pitot pressure decay of the manipulated jet C1 at $M_j = 1.83$. (a1) Natural jet, (a2) $\theta = 90^\circ$, (a3) $\theta = 120^\circ$, (a4) $\theta = 135^\circ$, (a5) $\theta = 150^\circ$.

the tab orientation, and this certainly has contributed to the differences in the potential core length. Third, the prominent number of shock-cell structures is reduced to 3 for all tab angles, as compared to 4 for the natural jet. Finally, now one may wonder whether the manipulated jet C2 also experiences jet bifurcation similar to that of C1. Thus, we present in Fig. 12 the typical shadowgraph images captured along the $x-y$ and $x-z$ planes of the manipulated jet C2 for $\theta = 90^\circ-150^\circ$ at $M_j = 1.83$. As explicitly seen in Figs. 12(a2)–12(b2), the tab placed in the major axis does not lead to jet bifurcation. Indeed, it is worth noting that tab placed along the major axis has a $h/a = 0.18$; thus, the tab may not induce a complete bifurcation of the manipulated jet. Thus, the centerline p_{0t}/p_{0s} is an authentic measure of jet mixing for the tabs placed along major axis (C2), unlike the tab placed along the minor axis (C1).

The mixing inferred from Fig. 11 is summarized by plotting the L_c^* with M_j in Fig. 13(a). The variation of L_c^* with M_j for $\theta = 90^\circ-150^\circ$ is presented in Fig. 13(a). In general, the mixing is reflected as the reduction in L_c^* . Shorter L_c^* signifies the higher jet mixing due to the tab. Once manipulated, the L_c^* for $\theta = 90^\circ-150^\circ$ is substantially reduced. As such, the L_c^* for the natural jet and the manipulated jet ($\theta = 90^\circ-150^\circ$) increases with an increase in M_j . Evidently, the L_c^* of $\theta = 135^\circ$ is the least among the manipulated jets, implying the maximized jet mixing. As L_c^* of the natural jet also increases with the increase in M_j , it would be interesting to interpret jet mixing of the manipulated jet with respect to the natural jet. Following Kumar *et al.*,^{23,24} we introduce another parameter, namely, percentage reduction in L_c^* , to quantify the mixing effectiveness of the manipulated jet as follows:

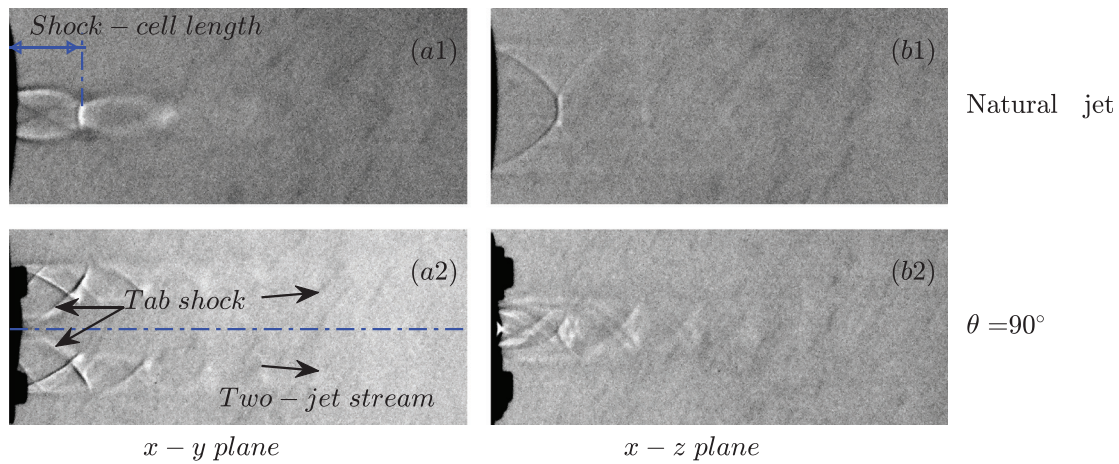


FIG. 10. Shadowgraph images of the natural and manipulated elliptic jet C1 captured along the $x-y$ and $x-z$ planes for $M_j = 1.83$; [(a1)–(b1)] Natural jet, [(a2)–(b2)] $\theta = 90^\circ$.

$$\Delta L_c^* = \frac{L_{c,Naturaljet}^* - L_{c,Manipulatedjet}^*}{L_{c,Naturaljet}^*}. \quad (9)$$

Larger the magnitude of ΔL_c^* , higher is the jet mixing due to the tab. As evident from Fig. 13(b), the ΔL_c^* of $\theta = 135^\circ$ is the maximum among the manipulated jets for all M_j , implying the maximized jet mixing. Unexpectedly, the results of the Mach 2.0 elliptic jet manipulated by triangular and rectangular tabs placed along the major axis^{7,14} reveal that the core length of the manipulated jet increases due to the control. That is, ΔL_c^* of the manipulated jet shows a negative sign (see Fig. 10 in Ref. 14 and Fig. 12 in Ref. 7). This highlights that the jet mixing of the manipulated jet decreases as compared to the natural jet. Thus, the observation reported in Refs. 7 and 14 is in distinct contrast to the present observation where the jet mixing of the manipulated jet increases due to the placement of tabs along the major axis. Apparently, the ratio h/b for the present investigation and that in Refs. 7 and 14 are 0.18 and 0.22, respectively. Further, the present investigation focused on the under-expanded state of the Mach 1.5 elliptic jet and Refs. 7 and 14 focused on the over-expanded state of the Mach 2.0 elliptic jet. Thus, a thorough and detailed study is required to address the in-differences. We hope to explain this in near future.

Now it would be interesting to compare the variation of ΔL_c^* with θ for any given M_j . Thus, we present in Fig. 14, the dependence of ΔL_c^* with θ for $M_j = 1.83$. It is worth noting that this cross plot is obtained from Fig. 13(b) for a given M_j . The ΔL_c^* decreases gradually to a minimum as θ is increased from 90° to 120° ; with further increase in θ to 135° , the ΔL_c^* rises sharply, reaching a maximum and then drops with further increase in θ to 150° . Evidently, the minimum and maximum ΔL_c^* are achieved for $\theta = 120^\circ$ and 135° , respectively. Thus, the optimized tab orientation angle θ for maximum jet mixing is found to be 135° .

It is worth correlating the geometric blockage for the tab angle $\theta = 90^\circ$ – 150° and its associated ΔL_c^* . Tabs at $\theta = 90^\circ$ offer the maximum geometrical blockage of 5% and the corresponding ΔL_c^* at $M_j = 1.83$ is about 20%. As θ increases, the geometric blockage reduces (Table I). As such, at identical $M_j = 1.83$, the maximum ΔL_c^* achieved for $\theta = 135^\circ$ is about 45%. Thus, at the most efficient orientation

angle, the blockage associated with tab is only about 3.35%. Hence, the thrust loss decreases due to the reduced geometric blockage. That is, lesser the tab blockage, lesser is the thrust loss associated with the tab.⁵⁴ Indeed, for $\theta = 150^\circ$, the ΔL_c^* achieved at $M_j = 0.83$ is about 40%, with the minimum tab blockage of 2.5%. Therefore, it may be stated that in addition to ensuring better mixing of $\theta \geq 135^\circ$, the thrust loss is also reduced. This suggests the superiority of tab orientation ($\theta \geq 135^\circ$) in comparison to placing the tab normal to the flow ($\theta = 90^\circ$).

The physical reason for the maximum jet mixing achieved at $\theta = 135^\circ$ is addressed here. As pointed out by Zaman *et al.*,¹⁷ there are two possible sources of streamwise vorticity for the flow over a tab:

- First is the pressure hill formed at the tab face, which, together with the presence of the wall, produces the pair of counter-rotating streamwise vortices.
- Second, owing to the pressure gradients on the front and rear surfaces of the tab, vortices are shed from the sides.

Initially, vortices shed by the tab have their axes parallel to the edge; as they proceed downstream, they get reoriented by the velocity gradients in the shear layer. Thus, if the tab is oriented in the upstream direction ($\theta < 90^\circ$), the primary and secondary source of vortices get canceled and would produce an outward bulge in the jet cross section (see Fig. 12 in Ref. 4). On the contrary, for $\theta \geq 90^\circ$, the primary and secondary sources of vortices get added and would result in an inward indentation;^{4,17,21,55} hence, the jet mixing increases.^{4,17} In other words, for $\theta \geq 90^\circ$, the vortices from primary and secondary sources add together, improving the tab effectiveness. Evidently, the main source of small-scale vortex generation is the pressure hill at the tab face and the vortices that are evolved from the sides of the tab. The pressure hill is formed due to the piling up of the fluid elements at the tab face. In a supersonic flow, as in the present case, a bow shock would be formed ahead of the tab [marked in Figs. 12(b3)–12(b5)]. Note that for $\theta = 90^\circ$, the shock that forms in the upstream direction of the tab is normal to the incoming flow direction. Hence, the formation of shock would result in a high-pressure zone downstream of the shock, resulting in the formation of a strong pressure hill over the tab face.

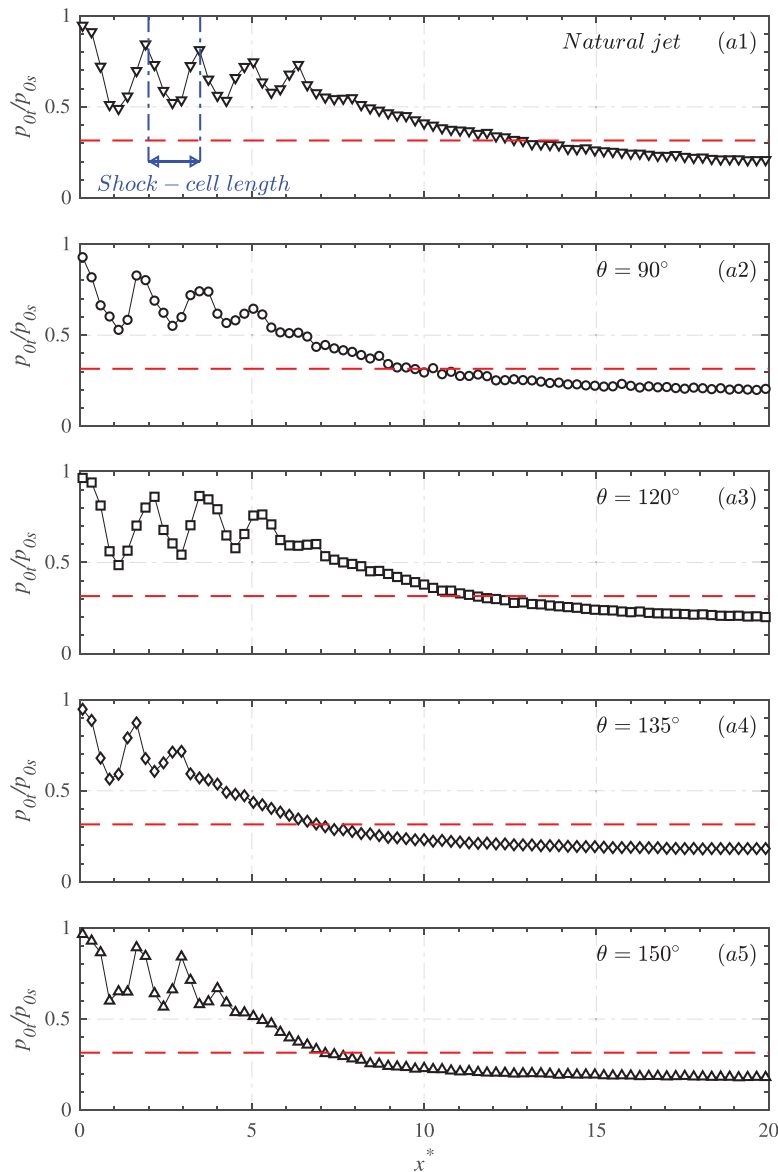


FIG. 11. Centerline Pitot pressure decay for the manipulated jet C2 at $M_j = 1.83$; (a1) natural jet, (a2) $\theta = 90^\circ$, (a3) $\theta = 120^\circ$, (a4) $\theta = 135^\circ$, (a5) $\theta = 150^\circ$.

However, for $\theta > 90^\circ$, the shock that forms in the upstream direction of the tab would be curved. Evidently, shock from both the tab intersects along the jet axis in the downstream direction and curves along the jet axis due to the momentum of the jet and crosses the jet axis as marked in Fig. 12(b3). This change in the shape of the shock would alter the shape of the pressure hill. Further, Foss and Zaman⁵⁵ investigated the optimum tab angle θ of a tab for increase mixing, by studying the peak streamwise vorticity magnitude ($|\Omega_{x-\max}|$) and its corresponding circulation. Interestingly, for all cases examined, the overall small-scale population was found to correlate directly with $|\Omega_{x-\max}|$. Interestingly, both overall small-scale population and $|\Omega_{x-\max}|$ quantities peaked at $\theta = 135^\circ$. Thus, the pressure hill at the face of the tab at $\theta = 135^\circ$ seems to establish the strongest $|\Omega_{x-\max}|$, which is the most efficient in mixing promotion, as reflected by the

shorter L_c^* or the maximized ΔL_c^* [Fig. 13(b)]. Now one may wonder why $\theta = 120^\circ$ has the least mixing as compared to $\theta = 90^\circ$. Indeed Foss and Zaman⁵⁵ argued that the peak magnitude of the corresponding circulation in the cross-sectional plane occurs for 90° and hence, ΔL_c^* was larger for $\theta = 90^\circ$ than $\theta = 120^\circ$.

It is well known that reduction in $\bar{L}_s^*$ and weakening of waves in the jet core can be taken as a measure of reduced shock-associated noise.^{5,49} Thus, we present in Fig. 14 the dependence of $\bar{L}_s^*$ with θ for $M_j = 1.83$. For reference the $\bar{L}_s^*$ of the natural jet at $M_j = 1.83$ is also presented in Fig. 14. A few observations can be made. First, the $\bar{L}_s^*$ of $\theta = 90^\circ$ and 120° exceeds that of natural jet, revealing the shock-cell length is marginally increased due to the control [Figs. 11(a2)–11(a3)]. On the other hand, the reduction in $\bar{L}_s^*$ is apparent for $\theta = 135^\circ$ and 150° as compared to the natural jet [Figs. 11(a4)–11(a5)]. This

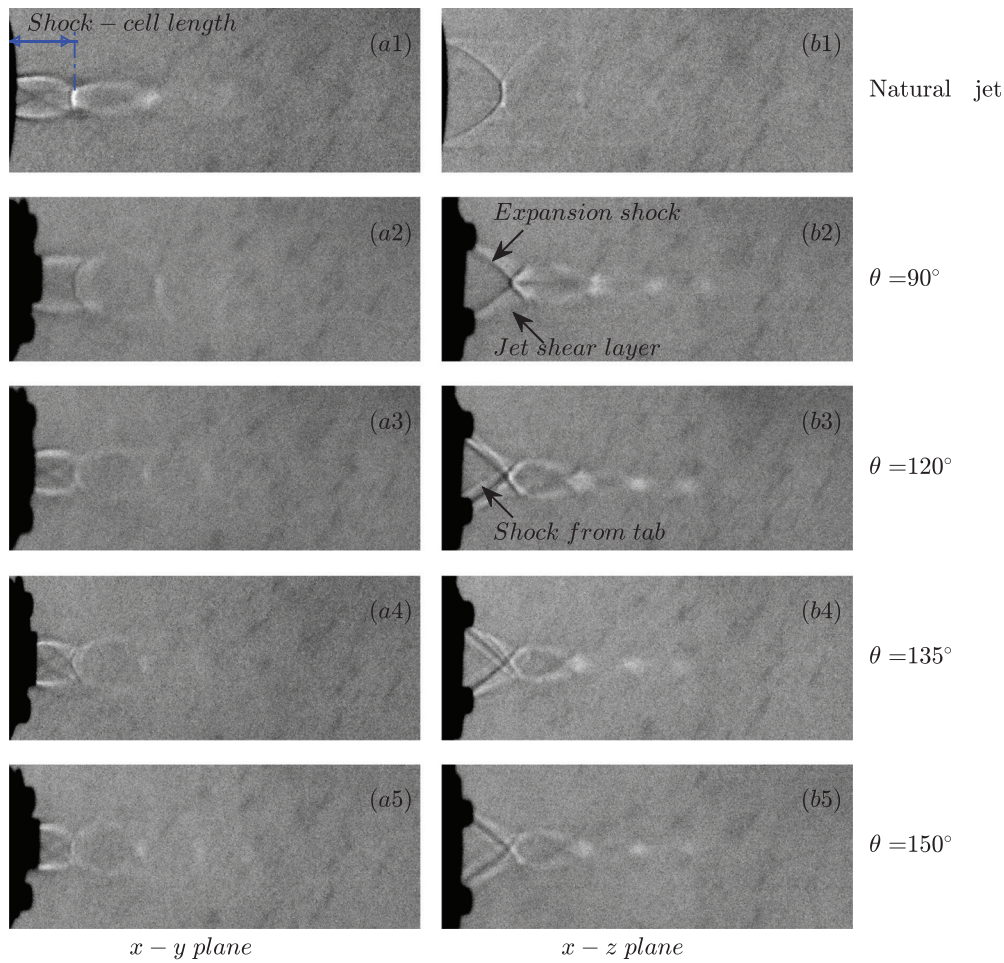


FIG. 12. Shadowgraph images of the natural and manipulated elliptic jet C2 captured along the $x - y$ and $x - z$ planes for $M_j = 1.83$; [(a1)–(b1)] natural jet, [(a2)–(b2)] $\theta = 90^\circ$, [(a3)–(b3)] $\theta = 120^\circ$, [(a4)–(b4)] $\theta = 135^\circ$, and [(a5)–(b5)] $\theta = 150^\circ$.

indicates a weaker shock-cell system in C2 at $\theta = 135^\circ$ and 150° relative to their natural counterparts, which is indeed necessary for overall jet noise reduction.^{29,49} Second, the maximum and minimum $\bar{L}_s^*$ of the manipulated jet corresponds to $\theta = 120^\circ$ and 135° , respectively. Coincidentally, the maximum ΔL_c^* and minimum $\bar{L}_s^*$ converges to $\theta = 135^\circ$. Similarly, the minimum ΔL_c^* and maximum $\bar{L}_s^*$ converges to $\theta = 120^\circ$. These observations reveal that ΔL_c^* and $\bar{L}_s^*$ are closely correlated with each other. Thus, Fig. 15 presents the dependence of ΔL_c^* with ΔL_s^* for $M_j = 1.83$ obtained from Fig. 14. As such, the ΔL_s^* is estimated based on the following:

$$\Delta L_s^* = \frac{\bar{L}_s^* \text{ Natural jet} - \bar{L}_s^* \text{ Manipulated jet}}{\bar{L}_s^* \text{ Natural jet}}. \quad (10)$$

It is very clear that, for a given M_j , ΔL_c^* is linearly correlated with ΔL_s^* . As expected, ΔL_s^* and ΔL_c^* exhibit a positive value for $\theta = 135^\circ$ and 150° (Fig. 15). Surprisingly, ΔL_s^* exhibits a negative value for $\theta = 90^\circ$ and 120° . On the other hand, ΔL_c^* exhibits a positive value for $\theta = 90^\circ$ and 120° (Fig. 15). Thus, it can be stated that an increased jet mixing

(ΔL_c^*) need not always signifies reduced strength of the waves (ΔL_s^*) in the jet core. Since weakening of the weakening of waves in the jet core can be taken as a measure of reduced shock- associated noise, the jets manipulated by tab $\theta = 90^\circ$ and 120° are an effective method to increase jet mixing but may not be effective in reducing the noise of the jet. Indeed, it has also been reported in open literature that tabs might also increase the noise of the jet and at the same time they would increase the jet mixing,⁵⁶ whereas the jet manipulated by tab $\theta = 135^\circ$ and 150° is an effective method to increase jet mixing and reducing the noise of the jet.

The variation of x_{as}^* with M_j for the manipulated jet C2 at $\theta = 135^\circ$ is shown in Fig. 16. For comparison, the variation of x_{as}^* with M_j of the natural jet is also shown. It is worth noting that x_{as}^* in Fig. 16 is obtained as explained in Sec. III. In general, x_{as}^* increases with M_j for both the natural and manipulated jets. Evidently, x_{as}^* for the manipulated jet C2 at 135° is lower than the natural jet, suggesting that the x_{as}^* moves in the upstream direction due to the placement of tabs. Indeed it is worth noting that the similar observation was also observed for other tab angles θ ; however, those results are not presented here. It is

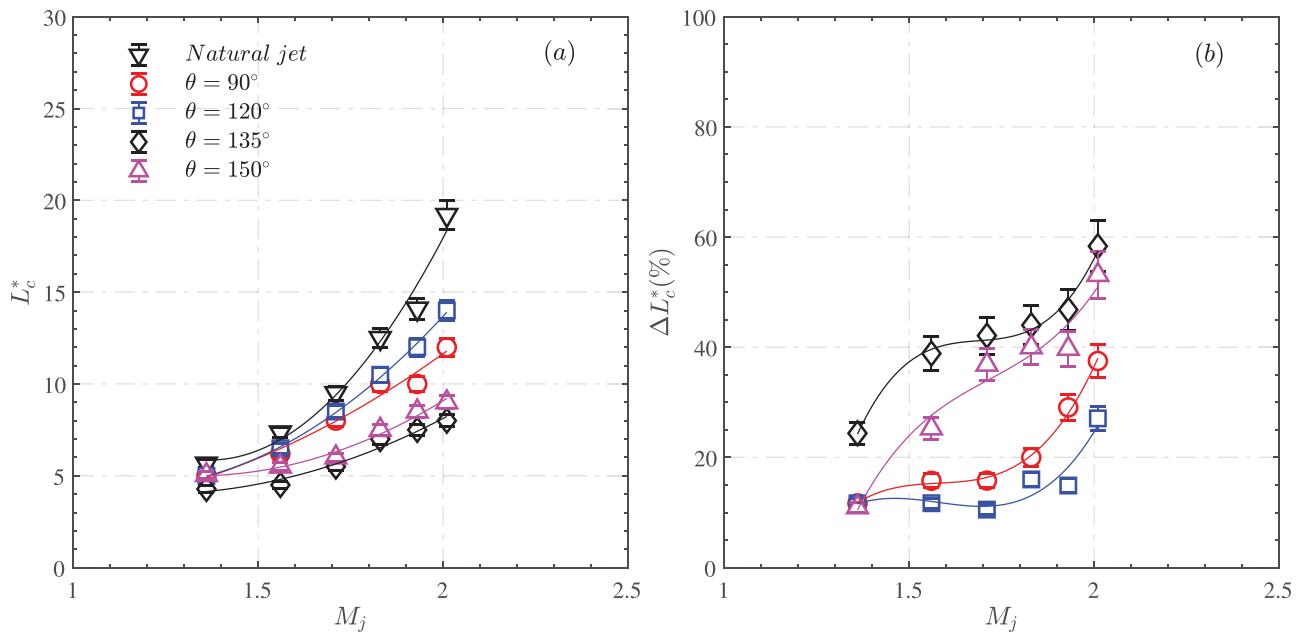


FIG. 13. (a) Dependence of L_c^* with M_j for the natural and manipulated jet C2 at $\theta = 90^\circ$ – 150° . (b) Dependence of ΔL_c^* with M_j for the manipulated jet C2 at $\theta = 90^\circ$ – 150° .

also well know that shifting the x_{as}^* in the upstream direction signifies increased mixing.^{7,13} Thus, the present study suggests that the jet mixing increases due to the placement of tabs. It is also worth commenting that axis-switching phenomena were not observed for the manipulated jet C1 for any given tab angle θ . Indeed, Zaman⁵⁴ and Kumar and Rathakrishnan¹⁴ also observed that axis-switching phenomena were

absent for the rectangular and elliptic jet manipulation via tab placed along the minor axis, respectively. Additionally, Zaman⁵⁴ and Kumar and Rathakrishnan¹⁴ also observed that the x_{as}^* moves in the upstream direction when the jets were manipulated by tab placed along the major axis. Thus, it has been found that the results obtained in this study are in good agreement with previous work.^{7,54}

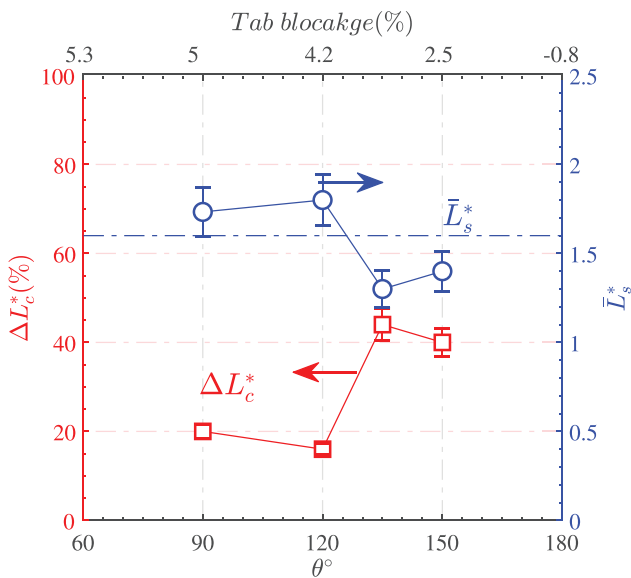


FIG. 14. Dependence of ΔL_c^* and L_s^* with θ for the manipulated jet C2 at $M_j = 1.83$. The dotted line here represents the L_s^* of the natural jet for $M_j = 1.83$ from Fig. 6.

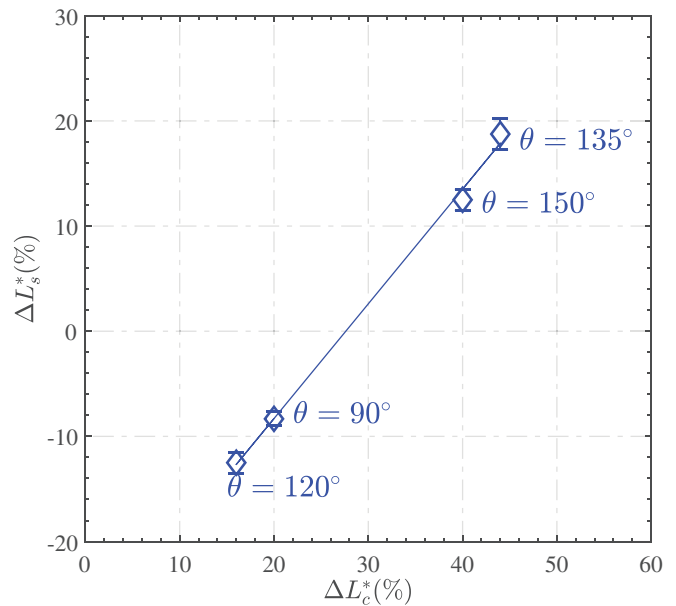


FIG. 15. Dependence of ΔL_s^* with ΔL_c^* obtained from Fig. 14 for the manipulated jets ($\theta = 90^\circ$ – 150°) at $M_j = 1.83$.

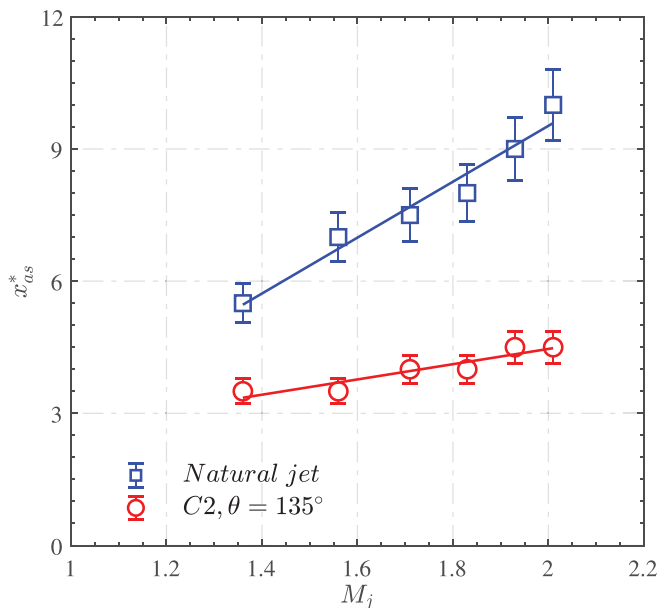


FIG. 16. Dependence of x_{as}^* with M_j for the natural and manipulated jet C2 for $\theta = 135^\circ$.

Zaman¹⁷ studied the effect of delta-tabs on mixing and axis switching of an AR3 elliptic nozzle at fully expanded jet Mach number of 1.67. Multiple tab configurations were studied, such as tabs employed in the minor, major, and at both minor and major axes of the elliptic jet. The jet with two delta-tabs placed along the major axis went through a rapid axis switching, whereas the jet with tabs along the minor axis continued to diverge on the major axis plane without switching its axes. Recently, Kumar and Rathakrishnan^{7,14} investigated in detail the effect of tab configuration on the mixing characteristics of a Mach 2 elliptic free jet at AR 2, 3, and 4. In their study, the tabs were employed separately at the minor and major axes of the elliptic jet. Their study revealed that the mixing promotion caused by the rectangular or delta-tabs employed along the minor axis results in a shorter core than the jet with tabs placed along the major axis and the uncontrolled jet. Interestingly, the tabs along the major axis retarded jet mixing at all the nozzle pressure ratios and hence, the core length of the manipulated jet becomes longer than the core of the uncontrolled jet itself.

V. CONCLUSIONS

A Mach 1.5 elliptic jet of aspect ratio 3.37 is experimentally investigated to understand the mixing promoting capability of the jet under manipulation. Two different tab configurations were studied, viz., tabs placed in minor (C1) and tabs placed in the major axis (C2) of the elliptic nozzle. For each tab configuration, there are four different slanting angles varying from 90° to 150° . Extensive Pitot measurements are performed along the jet centerline, and the jet stream has been visualized using the shadowgraph technique in the orthogonal planes of the jet. The normalized supersonic core length L_c^* , average shock-cell spacing $\bar{L}_s^*$, and axis-switching location x_{as}^* are examined to assess the jet mixing capability of the manipulated jets. The following conclusions are drawn from the investigation:

- (1) The supersonic core length L_c^* , shock-cell spacing $\bar{L}_s^*$, and the axis-switching location x_{as}^* for the natural elliptic jet increase as ideally expanded jet Mach number M_j increases. As a matter of fact, a close agreement on the dependence of $\bar{L}_s^*$ with M_j was observed with that of Tam's theory.⁴⁷ The superior mixing characteristics of an elliptic jet relative to a circular jet, which were found in previous works on the supersonic over-expanded jet (Kumar and Rathakrishnan¹²), prevail in the supersonic jet under-expanded jet as well.
- (2) The jet manipulation by C1 completely bifurcates the jet emulating the characteristics of a twin-jet, which is in distinct contrast to that Kumar and Rathakrishnan^{7,13} where the jet was completely bifurcated. Hence the centerline Pitot pressure measurement for bifurcated jet would over predict the jet mixing. As a matter of fact, the h/b ratio for the present investigation and Kumar and Rathakrishnan^{7,13} is about 0.43 and 0.62, respectively. Thus, employing the tab in the minor axis of the elliptic jet would be governed based on h/b ratio, where for $h/b < 0.5$, the tab may not induce a complete bifurcation of the manipulated jet.
- (3) For jet manipulation by C2, the mixing capability of the manipulated jet strongly depends on M_j and the tab orientation angle θ . As such the ΔL_c^* was linearly correlated to ΔL_s^* , revealing their dependence on jet mixing. As such, the maximum ΔL_c^* and ΔL_s^* was achieved for the jet manipulation by tabs oriented at $\theta = 135^\circ$, revealing the optimum tab angle for reducing noise and increased jet mixing, as also reported in open literature for a circular jet manipulated by tabs.⁴ When tabs are oriented in the downstream direction at $\theta = 135^\circ$, the pressure hill formed at the face along with the pressure gradient on the sides of the tab combine together to form a strong source of vortex generation. This leads to the small-scale population and $|\Omega_{x-\max}|$ quantities peaked at $\theta = 135^\circ$ and hence the maximum jet mixing of all configurations. On the other hand, the ΔL_s^* for $\theta = 90^\circ$ and 120° experiences a negative value while exhibiting a positive magnitude for ΔL_c^* but still less than that of $\theta = 135^\circ$. Presumably the tab angles $\theta = 90^\circ$ and 120° might increase the noise of the jet and at the same time would also increase the jet mixing. Finally, with suitable placement of the tabs C1 and C2, the axis-switching could be either stopped or augmented.

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DATA AVAILABILITY

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to ethical restrictions.

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